Parmida Davarmanesh

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MIT Ph.D. candidate in AI (LLM post-training, steering, interpretability) with first-author papers in PNAS and at a NeurIPS workshop; NSF Graduate Research Fellow and prize-winning speaker. Seeking full-time AI research scientist roles. **Available Aug–Sept 2027.**

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

- Ph.D., Computer Science (LLMs, mechanistic interpretability, AI alignment) — Expected 2027

Advisor: Dr. Ashia Wilson · NSF Graduate Research Fellow (2023–2026) · MIT EECS Distinguished Graduate Fellow (2021)

- M.S., Computer Science (thesis: 3D genome organization in 4-cell mouse embryos; advisor: Dr. Caroline Uhler) — 2023

University of Michigan, Ann Arbor, MI

- B.S., Mathematics (minor in Computer Science) · GPA 3.96/4.0 — 2021

PUBLICATIONS

1. **Parmida Davarmanesh**, Ashia Wilson, and Adityanarayanan Radhakrishnan. "Efficient and accurate steering of Large Language Models through attention-guided feature learning." Proc. Natl. Acad. Sci. U.S.A. 123(42): e2609978123 (2026).
2. **Parmida Davarmanesh**, et al. "Motif-aware Tokenization of the Genome: Towards Interpretable Modeling of Gene Regulation." *NeurIPS 2025 2nd Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences* (2025).
3. **Parmida Davarmanesh**, Qian Lin, Irene Tenison, Gabriel Jabbour, and Ridwan Alam. "Detection of Acute Myocardial Infarction Using Deep Learning on Lead-I ECG Data." *IEEE BSN* (2024), poster presentation.
4. **Parmida Davarmanesh**, Gabriel Jabbour, Qian Lin, Irene Tenison, Collin Stultz, and Ridwan Alam. "Detection of Acute Myocardial Infarction and Ischemia From Lead-I ECG Using Deep Learning." Circulation 150(Suppl 1): Abstract 4125281 (2024).
5. Shengpu Tang, **Parmida Davarmanesh**, Yanmeng Song, Danai Koutra, Michael W. Sjoding, and Jenna Wiens. "Democratizing EHR Analyses with FIDDLE: A Flexible Data-Driven Preprocessing Pipeline for Structured Clinical Data." JAMIA 27(12): 1921–1934 (2020).

EXPERIENCE

Massachusetts Institute of Technology, Cambridge, MA — 2025 – Present

Ph.D. Thesis Research: LLM Post-Training, Steering & Interpretability

- Raised the number of successfully steered concepts from under 50% to ~95% (512 concepts, Llama-3.1-8B), nearly doubling the prior state of the art with a computationally efficient method that scales to 70B-parameter models (PNAS, 2026).
- Currently post-training open-weight LLMs on mathematical reasoning with LoRA and GRPO, ORPO, and SFT objectives, as part of thesis work on recursive self-improvement with RLVR (reinforcement learning from verifiable rewards).

Thermo Fisher Scientific, Berlin, Germany — June – Aug 2024

Macromolecule Classification and Orientation Estimation in Cryo-ET Tomograms

- Trained 3D CNNs (Keras) on ~400,000 synthetic sub-tomograms: 0.99 per-class accuracy, 0.98 mean 6-DoF orientation correlation.
- Interpreted models with GradCAM++ and LRP; visualized 3D molecules, isosurfaces, and heatmaps in Amira.

MIT $100K Accelerate Entrepreneurship Competition — Sept 2025

Finalist and Audience Choice Award winner

- Pitched an ECG-based heart-attack detection device built on my IEEE BSN 2024 research.
- Engaged with cardiologists, patent attorneys, and angel investors about clinical utility and GTM strategy.

MIT Volunteer Consulting Group, Cambridge, MA

Project Leader — *Sep 2023 – Feb 2024*

- Framed indication expansion case with an early-stage VC-backed biotech company valued at $3.7M, recruited and led team of 6 consultants during 6-week engagement to produce a document strateg

Consultant (top 13% applicants) — *May – June 2023*

- Collaborated with a team of 6 consultants to shape customer relationship strategy for a science education non-profit via market research and interviews with 20+ key opinion leaders.

TEACHING

Teaching Assistant, Linear Algebra, MIT — Fall 2026

- Teach recitations, hold office hours, and build visuals and simulations that make abstract concepts tangible for students.

AWARDS

First Prize, ITA Graduation Day Talks (Information Theory & Applications Workshop) — 2026

Best Paper Award, MIT LIDS Student Conference — 2026

TECHNICAL TALKS

INFORMS, San Francisco, CA — Nov 2026

ML Tea Talk, MIT, Cambridge, MA — Oct 2026

MIT Statistics and Data Science Conference, Cambridge, MA — April 2026

Information Theory and Applications, San Diego, CA — Feb 2026

LIDS Student Conference, MIT, Cambridge, MA — Jan 2026